

Must Love Scrubs — Master Specification (v1)

The Everything Platform for Nurses

Single source of truth. Every page, question, AI response, and search result builds from this document. Supersedes scattered chat history; extends the design bible (`.claude/skills/mls-design-system`).

Status: Blueprint — confirm & outline phase. **We are NOT building the website or app yet.** We lock the architecture, then build a **web app** first, and only later a publishable (native) app.

0. What we are and aren't building right now

Building now: the *specification* — taxonomy, data model, content templates, relationship rules, question schema, AI rules, roadmap. The plan a developer/content team follows.

Build order (locked):

1. **Confirm & outline** the project (this document). ← we are here
2. **Web app** (the real product, database-backed).
3. **Publishable app** (native mobile) — later, plugging into the same APIs.

Explicitly deferred (do not build yet): native mobile app, the current static marketing site as the product, simulation/virtual hospital, 3D models, live tutoring, career center. These stay on the roadmap; they plug into the same ecosystem when their time comes.

The current static site (the clickable preview we built) remains a **marketing/prototype front** — useful for showing the vision. The web app is a separate, database-backed build.

1. Master Vision

Mission: Be the operating system for nursing education — not another NCLEX site, but the connected knowledge engine nurses use from school through licensure and into practice.

Positioning: Netflix × Duolingo × MasterClass for nursing, built "for nurses, by nurses." Competitors (Bootcamp, UWorld, nursing.com, Archer, Kaplan, Hurst) sell question banks + videos. Our moat is the **Nursing Knowledge**

Engine: every concept connected, so learning is a *guided path*, not a pile of pages.

Durable advantage: built like a software company — schema, API contracts, content standards, and AI rules defined *before* mass content. New features plug into the same ecosystem instead of being rebuilt.

Launch milestone (not a ceiling): 5,000–8,000 questions. Architecture must scale far beyond without redesign. Value = quality + connections + personalization, not raw count.

2. Phase 0 — Platform Architecture

The core: Nursing Knowledge Engine (NKE)

One central brain. Every object (disease, drug, lab, procedure, symptom, skill, question, video, flashcard) is a **node**; every relationship is an **edge**. A single concept (e.g., Heart Failure) links to its symptoms, meds, labs, interventions, procedures, videos, images, questions, NGN cases, flashcards, memory tricks, Esi explanations, specialties, anatomy, and patient education.

Product ecosystem (modules that ride on the NKE)

- **Learning:** Nurse/Scrubs Dictionary, Disease Encyclopedia, Medication Center, Lab Center, Procedure Library, Clinical Skills Library, Anatomy & Physiology.
- **Practice:** Question Engine (all NGN item types), NGN Case Studies, Flashcards (spaced repetition), Smart Study Engine (daily plan), Memory Engine.
- **Esi (AI):** tutor / coach / study partner / mentor — personalized to the student's data.
- **Dashboard:** readiness, pass probability, weak/strong areas, streaks, per-domain scores.
- **ScrubTV:** media network (videos, channels, rapid reviews) — *watch only; no user video uploads*.
- **Community:** Twitter-style feeds + threads for nurses and visitors (see §7).
- **Later (deferred):** Simulation Center, Career Center, native mobile app.

AI strategy (Esi)

Esi answers from **structured NKE content first**, then explains in plain language. She is personalized (knows score, weak topics, history, specialty, exam date, confidence), explains *reasoning* not just answers, offers follow-up practice, and links related concepts. Guardrails from the design bible remain: educational tutor only, never diagnoses/prescribes, paywalled (no free open chat).

3. Phase 1 — Master Content Architecture

3.1 Content types (codes)

NKI topic · **DIS** disease · **MED** medication · **LAB** lab · **PRO** procedure · **SKL** clinical skill · **ANA** anatomy/physiology · **SYM** symptom · **ASM** assessment · **QBK** question · **NGN** case study · **FLS** flashcard · **VID** video · **IMG** image/diagram · **EDU** patient education · **PATH** learning path.

3.2 Taxonomy tags (every item is tagged)

- **Body system:** Cardiovascular · Respiratory · Neurological · Renal/Urologic · Endocrine · Gastrointestinal · Hematologic · Immune/Infectious · Musculoskeletal · Integumentary · Reproductive/OB · Neonatal · Pediatric · Psychiatric.

- **Specialty:** the 22 any-RN specialties (design bible list).
- **NCLEX Client Needs:** **SMS** Safe & Effective Care · **HPM** Health Promotion · **PSY** Psychosocial · **BPI** Basic Care & Comfort · **PHA** Pharmacological · **RRT** Reduction of Risk · **PAD** Physiological Adaptation.
- **Clinical Judgment (NGN):** **REC** Recognize Cues · **ANA** Analyze Cues · **PRI** Prioritize Hypotheses · **GEN** Generate Solutions · **ACT** Take Action · **EVA** Evaluate Outcomes.
- **Difficulty (1–5):** 1 Foundational · 2 Basic Application · 3 Clinical Application · 4 Prioritization · 5 Complex NGN/Critical Thinking. (*Maps to the qbank engine's numeric level*.)
- **Age group:** Adult · Pediatric · Neonatal · Geriatric. **Care setting:** ER · ICU · Med-Surg · L&D · Pediatrics · Community · LTC.
- **Exam:** RN · PN · Specialty. **Bloom level.** **Estimated time.**

3.3 ID convention

TYPE-SYSTEM-#### → e.g. **DIS-CV-0001** (Heart Failure), **MED-CV-0007** (Furosemide), **LAB-CV-0002** (BNP), **QBK-CV-0001**. Stable, unique, human-readable. De-dup key.

3.4 Content templates (locked)

Each content type has a fixed template so thousands of entries stay consistent. Summaries below; full field lists in §5 of the design bible + this doc's appendix intent.

- **Disease:** definition · causes · risk factors · pathophysiology · signs/symptoms · assessment · diagnostics · lab findings · nursing diagnoses · interventions · medications · patient teaching · complications · priority actions · NCLEX tips · related questions/videos/terms.

- **Medication:** identity (generic/brand/classes/pronunciation) · overview · MOA · uses · dosage/routes · contraindications · warnings (incl. black box) · side effects (common/serious/life-threatening) · nursing considerations (pre/during/post) · patient education · related content.

- **Lab:** identity · normal & critical values (adult/peds) · purpose · high causes · low causes · nursing actions · notification thresholds · patient education · related content.

- **Procedure:** identity/setting · purpose/indications · equipment · preparation · steps · risks/complications · nursing responsibilities (before/during/after) · patient teaching · documentation · related content.

- **Clinical Skill / Anatomy / Flashcard / Video / Image / Patient Education:** each has its own locked template (see §6 modules of prior spec).

3.5 Relationship engine

Every item declares typed links to related items. Powers "related content," Clinical Connections sidebar, adaptive review, and search. Example — **Hyperkalemia** auto-links: diseases (AKI, CKD, Addison's), meds (spironolactone, lisinopril, K⁺ supplements), labs (K⁺, BUN, creatinine), procedures (dialysis, ECG monitoring), questions & NGN cases.

3.6 Search & index rules

Any query (synonym, abbreviation, brand/generic) returns a **unified result**: primary page + related meds/labs/procedures + NCLEX questions + NGN cases + flashcards + Esi explanation + videos. Example:

"Lasix" → Furosemide (primary) + Heart Failure/Pulmonary Edema + K^+/Na^+ + IV administration + 12 questions + 2 cases.

3.7 AI integration rules

Esi: structured content first → plain-language explanation → matches level → explains reasoning → offers practice → links related topics. Every AI answer is traceable to NKE nodes.

4. Phase 2 — Nursing Knowledge Index (the graph)

4.1 Node & edge model

Nodes = content objects (§3.1). Edges = typed relationships (has-symptom, treated-by, monitored-with, causes, complication-of, tested-by, teaches, assesses, links-to). This is the "Google Maps of nursing knowledge."

4.2 Adaptive learning via relationships (the secret weapon)

A wrong answer is not just "incorrect." Missing a hyperkalemia question raises priority on the *connected* nodes: potassium physiology, ECG changes, kidney disease, ACE inhibitors, spironolactone, dialysis, arrhythmias, related labs, fluid/electrolyte disorders. The system builds a **personalized learning path** from the graph, not just a score.

4.3 Student knowledge graph

Each student has a dynamic map: per-domain mastery (Cardiovascular 92%, Pharmacology 61%, Labs 54%, NGN 48%, Med Calc 35%...), a forgetting-curve per node, confidence scores, and history. Esi's memory reads this ("you always miss insulin," "haven't reviewed burns in 18 days," "confidence dropping in electrolytes").

4.4 Signature UX features

- **Clinical Connections panel** (sidebar): "if you understand this, learn next..." — a guided path.
- **"Why?" chain**: tap Why repeatedly to drill from a finding down through physiology, each step a deeper NKE link.
- **Concept maps**: interactive visual branches (e.g., Sepsis → infection, lactate, cultures, antibiotics, fluids, vasopressors, organ dysfunction, shock).

4.5 CAT ↔ pop quizzes & mini-prep ↔ visible improvement (locked)

This is one closed loop, not separate features:

1. **CAT / readiness exam** estimates the student's ability and pinpoints weak nodes (by system, topic, and clinical-judgment step).
2. Those weak nodes are pushed into **pop quizzes** (short, adaptive, 5–10 items that surface at smart moments) and **mini-prep** (focused micro-lessons + drills on exactly those gaps).
3. Every attempt updates the **student knowledge graph** (mastery + spaced-repetition schedule), which **re-feeds the next CAT** — so the adaptive test, the pop quizzes, and the mini-prep are all driven by the *same* mastery data.

4. **Improvement is shown, not implied:** the dashboard tracks each domain's mastery **over time** (before → after), readiness trend (Low → Very High), a per-topic "then vs now," and pop-quiz score deltas — so the student literally sees weak areas turning into strong ones.

Is this how it works already? The *architecture* already supports it — the design uses one shared mastery store that feeds `/me/next` (smart study) and the dashboard. What this update does is make the loop **explicit and named**: CAT results must generate targeted pop quizzes + mini-prep, and the dashboard must show the before/after improvement, not just a current score.

5. Data model & schema (build-like-a-software-company)

Relational core + a graph/edge table for relationships. Illustrative — the developer finalizes types, but these contracts are locked in intent.

nodes — `id` (TYPE-SYS-####, PK) · `type` · `title` · `body_system` · `specialties[]` · `difficulty` (1–5) · `client_needs[]` · `cj_steps[]` · `age_groups[]` · `care_settings[]` · `status` (draft/reviewed/published) · `version` · `updated_at` · `body` (structured JSON per template) · `search_terms[]`.

edges — `from_id` · `to_id` · `relation` (typed) · `weight`. (Directional; powers related-content + adaptive priority.)

questions (extends node) — `type` (mc/sata/matrix/bowtie/dropdown/ordered/hotspot/trend/audio/image/casestudy) · `stem` · `options[]` (each {`text`, `rationale`}) · `correct` (index | list | per-part) · `tip` · `pearl` · `references[]` (internal QC) · `learning_objective` · `bloom` · `est_time` · `links[]` (diseases/meds/labs/procedures) · plus all node tags.

Rationale required for every option (validator-enforced).

users · **attempts** (user_id, question_id, chosen, correct, ms, ts) · **mastery** (user_id, node_id, score, confidence, last_seen, next_review) · **streaks** · **entitlements** (plan, expiry, esi_included).

posts / **threads** (community — §7).

Content standards: 100% original; QC gate per item (verified answer, NGN wording, no ambiguity, rationale per option, difficulty tagged, dedup, correct placement, references logged). Free/paid gating is a data flag (`free`), not code.

API contracts (intent): `GET /node/:id` (with related), `GET /search?q=`, `GET /questions?filters`, `POST /attempt`, `GET /me/dashboard`, `GET /me/next` (smart study), `POST /esi/ask` (returns answer + cited node ids), community `GET/POST /threads`. Same APIs serve web app now and native app later.

6. Question Engine — ALL NCLEX + NGN item types (verified vs NCSBN)

We support the **complete** set. NCSBN uses **2 stand-alone** NGN item types (bow-tie, trend) plus **12 case-study** NGN item types, alongside the **traditional** item types that remain on the exam. Our engine must cover every one.

Traditional NCLEX item types

1. **Multiple Choice** (single answer) — *built*
2. **Multiple Response / Select All That Apply (SATA)** — *built*
3. **Fill-in-the-Blank / Dosage Calculation** (numeric entry) — planned
4. **Ordered Response** (drag-to-rank) — planned
5. **Hot Spot** (click the region on an image) — planned
6. **Chart / Exhibit** (tabbed EHR + question) — planned
7. **Graphic Options** (answer choices are images) — planned
8. **Audio** (listen, then answer) — planned

NGN stand-alone item types

9. **Bow-Tie** (condition + actions + parameters) — *built*
10. **Trend** (data across multiple time points) — planned

NGN case-study item types (the 12, as families + variants)

11. **Matrix** — Matrix Multiple Choice (*built*) + Matrix Multiple Response (multi-pick per row) — planned variant
 12. **Extended Multiple Response** — SATA / **Select-N / Grouping**, with **partial credit** — planned
 13. **Cloze / Drop-Down** — drop-down, drop-down cloze, drop-down table, drop-down **rationale** — planned
 14. **Extended Drag-and-Drop** — incl. **drag-and-drop cloze** (not every token must be used) — planned
 15. **Enhanced Hot Spot / Highlight** — highlight **text** or highlight **table** — planned
- + **Unfolding Case Study** — the wrapper: **6 items** stepping through the 6 clinical-judgment stages (Recognize → Analyze → Prioritize → Generate → Take Action → Evaluate).

Built today (qbank engine): MC, SATA, Matrix (multiple-choice), Bow-Tie. Everything else is specified and planned; each gets its own interface + scoring, added one at a time.

Every question (all types): rationale for right *and* wrong, difficulty (1–5), topic, specialty, NCLEX Client-Needs category, clinical-judgment step, est. time, references, and NKE links. Partial-credit scoring where NCSBN uses it (SATA/extended multiple response, matrix, bow-tie). CAT-ready: difficulty drives adaptive item selection (IRT/Rasch-style; recalibrated from real **attempts** data).

7. Community — Twitter-style for nursing (locked decision)

- **Format: feeds + threads**, full X/Twitter-style, for **nurses and visitors** alike.
- **Full media posting** — **text, images, video, and GIFs** (whatever you can do on Twitter/X). Users can post video here. (*ScrubTV stays a separate, curated MLS media network; the community is user-generated.*)
- **Profiles have NO feed** — profiles stay focused on learning/dashboard, not a social wall.
- **Keep the learning dashboard as-is for now.**
- Purpose: discussion, tips, encouragement, study-group energy — the community moat competitors' closed Facebook groups can't match, native to the platform.

- Moderation & safety required for user media (report/flag, content rules) since video/images are allowed.
 - Roadmap extras (later): study groups, leaderboards, weekly challenges, mentors.
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8. Gold Standard Module #001 — Heart Failure

The proven template for every future topic. Full module authored ([DIS-CV-0001](#)): Quick Learn · pathophysiology map · left/right types with memory hooks · assessment · labs (BNP, troponin, electrolytes, creatinine) · medication connections (furosemide, lisinopril, metoprolol, spironolactone, digoxin) · priority nursing actions · patient education · "PUMP" memory system · linked NCLEX question · NGN case (recognize → analyze → take action) · connected knowledge index · Esi training prompt. **Standard = build one gold module end-to-end, approve it, then replicate.** Next gold module: **Diabetes Mellitus** (tests endocrine + meds + labs + calculations + complications + NGN).

9. Development roadmap

1. **Confirm this spec** (now).
2. **Foundation build:** database (nodes/edges/questions/users/mastery), API contracts, auth, content-ingest pipeline (extends the existing [data/qbank](#) batch pipeline — same schema philosophy).
3. **Web app v1:** Dictionary + Disease/Med/Lab/Procedure pages (NKE-linked) · Question Engine (all types) · Dashboard · Esi (structured-first) · Smart Study · Community (feeds/threads).
4. **Content at scale:** gold modules → replicate; question batches toward 5–8k (benchmark-first, QC gate).
5. **Native app:** wraps the same APIs.

Automate vs. human: AI drafts content/questions in batches → **human (nurse SME) review before publish** (status flag). Never ship unreviewed clinical content.

10. Build instructions & standards (for Claude / developers)

- **Software-company discipline:** schema + API contracts + content standards + AI rules are locked *before* mass content. Features plug into the NKE; nothing is rebuilt per feature.
- **IDs & templates are mandatory** — no free-form pages. Every item validates against its template + tag set (extend [tools/build_qbank.py](#) -style validators to all content types).
- **Relationships are first-class** — every item ships with its edges; "related content" and adaptive learning depend on them.
- **Original content only**, QC gate enforced, references logged internally.

- **UI principles:** mobile-first, the locked Violet & Mint design system, accessible, fast. Esi paywalled, educational-only disclaimer.
 - **Data-driven gating** (free/paid) — flags, not forks.
 - **Web app before native app;** both consume identical APIs.
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11. Locked decisions captured from this session

- Do **not** build the website or app now — confirm & outline first.
 - **Web app first**, publishable (native) app later.
 - Community = **full Twitter-style feeds + threads** (text, images, **video**, GIFs), nurses **and** visitors.
 - **Profiles: no feed. Dashboard: keep learning focus for now.**
 - **CAT ↔ pop quizzes ↔ mini-prep** is one loop on shared mastery data; dashboard shows **improvement over time** (§4.5).
 - **All NCLEX + NGN item types** are in scope (§6), verified against NCSBN.
 - Everything hangs off the **Nursing Knowledge Engine** (connected nodes + edges).
 - Gold-standard-module approach: perfect one, then replicate (HF done; Diabetes next).
 - Launch target 5–8k questions; architecture scales beyond without redesign.
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End of Master Specification v1. Next step on approval: turn §5 (data model) + §6 (question engine) into the concrete database + API contract document, then scaffold the web app foundation.